

MUHAMMAD AWAIS

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **3+ years** of experience building **scalable web applications, cloud-native solutions**, and **AI-driven platforms**. Expertise in MEAN/MERN stack, cloud deployments, and real-time communication systems. Adept at leading end-to-end development, integrating third-party APIs, and delivering high-impact software products.

TECHNICAL SKILLS

Domain	Technologies
Languages	JavaScript, TypeScript, Python
Frontend	React, Redux Toolkit / RTK Query, Angular, Tailwind CSS, Bootstrap, MUI
Backend	Node.js, Express.js, REST APIs, Socket.IO
Databases	MongoDB, Firebase, Redis
AI & Communication	OpenAI LLM, Twilio, Deepgram STT/TTS, Google STT/TTS, Infobip
Cloud & DevOps	AWS (EC2, S3, CloudFront), Google Cloud Run, Docker, Kubernetes, Nginx, CI/CD
Tools & Practices	Git, Bitbucket, Jest, Postman, Jira, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Callvio

📅 May 2025 – Present | 🔗 [callvio.co](#) • [Dashboard](#)

Led development of an AI-powered Receptionist App for automating business customer interactions through intelligent voice assistance.

- Architected a **full-stack AI receptionist platform** handling live customer calls with natural language understanding and automated responses
- Engineered a responsive **React frontend** with **Redux Toolkit** for state management, **RTK Query** for efficient API caching, and custom middleware for analytics and logging
- Designed scalable **Node.js microservices** integrating **Twilio** for voice routing, **Deepgram & Google STT/TTS** for real-time speech processing, and **OpenAI LLM** for conversation intelligence
- Implemented **Infobip WebSocket** integration for real-time caller speech capture and Jazz telephony support

- Built intelligent **appointment booking**, business FAQ handling, and real-time automated response systems
- Set up **CI/CD pipelines** with automated testing; deployed backend on **Google Cloud Run** for auto-scaling
- Integrated **Redis** for session management, caching, and real-time data processing ensuring low-latency responses
- Combined telephony, AI, and cloud-native technologies, significantly improving **customer engagement and operational efficiency**

Tech: React Redux Toolkit RTK Query Node.js Firebase Redis Twilio OpenAI Deepgram Google TTS/STT Infobip Cloud Run Vercel

Senior Software Engineer — Binary Tech (*DoctorNow*)

 May 2022 – April 2025 |  docnow.ai • [Live Dashboard](#)

Developed and maintained core features for DoctorNow — a comprehensive telemedicine platform with EHR, real-time consultations, and digital prescriptions.

- Led **front-end architecture** using Angular with lazy loading, AOT compilation, and optimized bundle sizes for desktop and mobile
- Engineered scalable **backend APIs** using Node.js and Express with modular architecture, middleware pipelines, and comprehensive error handling
- Designed and optimized **MongoDB schemas** with indexing, aggregation pipelines, and query optimization for managing large volumes of patient data
- Built **real-time chat** modules using Socket.IO for doctor-patient communication during consultations
- Implemented secure **authentication flows** using JWT, refresh tokens, HTTPS enforcement, rate limiting, and input validation/sanitization
- Integrated **Twilio** (SMS/voice), **Zoom SDK** (video consultations), and **payment gateways** for seamless third-party services
- Developed **digital prescriptions**, video consultation rooms, appointment scheduling, and patient history tracking
- **Reduced page load time by 40%** and **increased user engagement by 25%** through code optimization, CDN integration, and UI revamp
- Led **Agile ceremonies** — sprint planning, code reviews, mentoring junior developers, and CI/CD pipeline management

Tech: Angular Node.js Express.js MongoDB Socket.IO JWT Twilio Zoom SDK AWS Docker Jest



EDUCATION

Bachelor of Science in Computer Science | 2018 – 2022 *Punjab University College of Information Technology (PUCIT), Lahore, Pakistan*